Introduction to HTML and CSS

→ HTML (Hyper Text Markup Language) is the standard markup language used to create and structure content on web pages. It defines the basic structure of a webpage using elements and tags such as headings, paragraphs, links, images, lists, tables, forms and other multimedia content.

→ CSS (Cascading Style Sheets) is a stylesheet language used to control the presentation and visual appearance of HTML documents. It is used to define properties such as colors, fonts, backgrounds, spacing, borders, alignment, dimensions, and layouts. CSS also supports responsive design, allowing webpages to adapt to different screen sizes and devices. CSS can be applied using inline, internal, (or) external style sheets.

HTML Components:-

1) HTML Tags - Define the structure and elements of a webpage.

2) Headings - used to display titles and headings using <h1> to <h6>

3) Paragraphs - used to display text using the <p> tag

4) Links - used to navigate between webpages using <a> tag.

5) Images - used to display images using <image> tag.

6) Lists - used to Create ordered and unordered lists using <ol>, <ul>,

7) Tables - used to organize data in rows and columns.

8) Forms - used to collect user input using elements such as text boxes, buttons and checkboxes.

9) Div and Span - used to group and organise webpage content.

10) Semantic elements - Elements such as <header>, <nav>, <section>, <article>, <footer> Provide meaningful webpage structure.

CSS Components:-

1. Selectors:- Select HTML elements to style them.
2. Properties - Define the visual characteristics of elements.
3. Values - Specifies the setting of a CSS property.
4. Colors - Define text, background, border colors.
5. Fonts - Control font family, size, style and weight.
6. Backgrounds - Set background colors and images.
7. Borders - Define the border style, width and radius.
8. Margin - Controls the space outside an element.
9. Padding - Controls the space inside element.
10. Layout - Controls the arrangement of elements using Flexbox, Grid and Grid and positioning.

Experiment 05

05) Implement Modules in NodeJs

In Nodejs, you can implement modules using two distinct systems : CommonJs(CJS), which is traditional default, and Esmodule (ESM), the modern JavaScript standard.

① The CommonJs Method (default):

CommonJs uses Module.exports to share code and require() to import it. This system is modules synchronously.

a) Create the module (math.js)

```
Const add = (a,b) => a+b;
Const Subtract = (a,b) => a-b;
Module.exports {
    add,
    Subtract
};
```

b) Use the module (app.js):

```
Const math = require('./math');
Console.log(math.add(10,5));
Console.log(math.Subtract(10,5));
```

Output: 15
5

Experiment-04

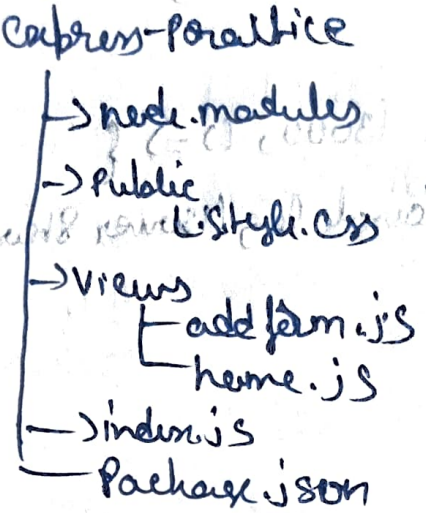
(4) Develop dynamic webpage using Java Script

Program:

```
<!DOCTYPE html>
<html>
<head>
    <title> Dynamic webpage </title>
    <style>
        body{
            text-align : Center;
            Font-family: Arial;
            margin-top: 100px;
        }
    button {
        Padding : 10px 20px;
        Font-size: 16px;
        Cursor : Pointer
        }
    </style>
    </head>
    <body>
        <h1 id : "heading"> welcome to My
                            webpage </h1>
        <p id = 'message'> Click the button to
                           Change this webpage</p>
        <button onclick = "changeText()"> Click Me
                                          </button>
```

Experiment-No6:- Develop a mini application using Node.js and Express.JS

Folder Structure:



Code:-

→ index.js

```
Const express = require ('express');
const app = express();
const bodyParser = require ('body-parser');
app.set('view engine', 'ejs');
app.use (express.static('Public'));
app.use(bodyParser.urlencoded({extended: true}));
let recipies = [];
app.get ('/', (req, res) => {
        res.render ('home', {recipies});
})
app.get ('/add', (req, res) => {
        res.render ('addform');
})
```

Experiment-07

7) Create a Functional Component and class Component using React.

Aim:- TO develop a react application using Functional Components and class components to display student details, manage State, handle button clicks and dynamically update data on a web page.

- Functional Component:

A Functional Component is a Javascript function that returns JSX to display the user interface. It uses React Hooks such as useState to manage State and update the webpage dynamically.

Program:-

```
import React, {useState} from
'react';

function StudentFunctional(){

    const [name, setName] = useState("Rakesh");
    const[course, setCourse] = useState("CSE");
    const[count,setCount] = useState(0);

    const changeName = () => {

        setName("Ravi");
```

Experiment - 08

8) Implement Http methods using JSNOD Server.

Aim:- To implement HTTP methods such as GET, POST, PUT, PATCH and DELETE using JSON server to perform CRUD (Create, Read, Update and Delete) operations on student records.

Theory:- JSON Server is a tool used to create a fake REST API using a JSON file. It allows us to perform HTTP operations without developing a backend server from scratch.

- HTTP methods are used to communicate between a client and a server.
- GET - Retrieves data from the server.
- POST - Adds new data to the server
- PUT - Replaces an existing record
- PATCH - Updates selected fields of a record
- DELETE - Removes a record from the server.

Experiment-9

09) Develop CRUD operations using MongoDB.

Aim:- To develop a web application using Node.js, Express.js and MongoDB to perform CRUD operations (Create, read, update, delete) on student records.

Theory:- MongoDB is a Non-sql database that stores data in flexible, JSON like documents, CRUD operations are used to manage data in a database.

Mongoose is an Object Data Modeling (ODM) library for MongoDB and Node.js It helps define schemas, interact with collections and perform database operations.

1) Installation and Project Setup.

```
mkdir MongoDB-CRUD
cd MongoDB-CRUD
npm init -y
npm install express mongoose
```

Program:- Server.js